CI2-221 Computational Methods II

Coursework – Modelling of a Sway Structure

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1.0 Introduction

The aim of the coursework is to find the most efficient solution to a transversely-loaded sway steel structure.

1.1 Matlab scripts

There are in total six m files, namely: Coursework, TRUSS, FRAME, TrussSway, FrameSway and TrussOptimisation. All dimensions used are in N and mm. 'Coursework' is the main script used to solve the questions; it contains seven sections. Based on the structure given and using the number of bays and storeys, the first section (called 'Initialization') generates the nodal coordinates, the element connectivity, free and restrained degrees of freedom (dofs), the restrained displacements, the forces on the structure and the possible position of the cross members on a storey. The five sections that follow solve questions one to six, which will be explained in details, while the last section further optimises the truss structure.

'TRUSS' and 'FRAME' are classdef files. Generalization is the focus, i.e. they can be used to solve any truss and frame structure given sufficient inputs. Therefore, the code that generates of node coordinates and element connectivity specific to this structure are written in the main script. The methods and their purposes are to:

TRUSS or **FRAME**:	Create a variable with the class, calculate EA and EI of each element, determine the number of nodes and elements, and define the numbering of dofs.
Volume:	Calculate the total volume of material used in the structure.
Solve:	Construct stiffness matrix K, initiate displacements U and forces F, solve for unknown dofs and reactions, update nodal coordinates.

MaximumLateralSway:	Determine the maximum lateral sway.
Condition (truss only):	Calculate the value of rcond of the matrix K_{FF}.
Plot:	Plot the structure with the amplitude given, if the axial loads have been calculated (using the method AxialLoad), then they are shown on the figure too.
Display:	Display the unknown displacements and reactions.
EquilibriumCheck:	Create a two-element vector 'check', output '1' if the check is passed and '0' otherwise. The two elements correspond to the horizontal and vertical direction respectively.
DisplayEquilibriumCheck:	Perform an equilibrium check and display the results.
AxialLoad:	Calculate the axial load in each element using Hooke's Law.
BucklingCheck:	Calculate Pcr (Euler buckling) as well as the margin of safety of each element.
PlotBuckling:	Create a figure that shows the elements at risk of buckling.
BucklingRadius (truss only):	Find the radius such that the margin of safety is zero.
PlotRadius (truss only):	Create a figure that shows the radius of each element.

'TrussSway', 'FrameSway' and 'TrussOptimisation' are functions that are used to find the radius required to obtain 25mm of lateral sway. This is further discussed in section 5.0 and 7.0.

1.2 Node and element numbering convention

The main script ('Coursework') can be run with any number of bays and storeys. In this report, a structure with three bays and five storeys is analysed. The numbering convention used for nodes and elements are as follows:

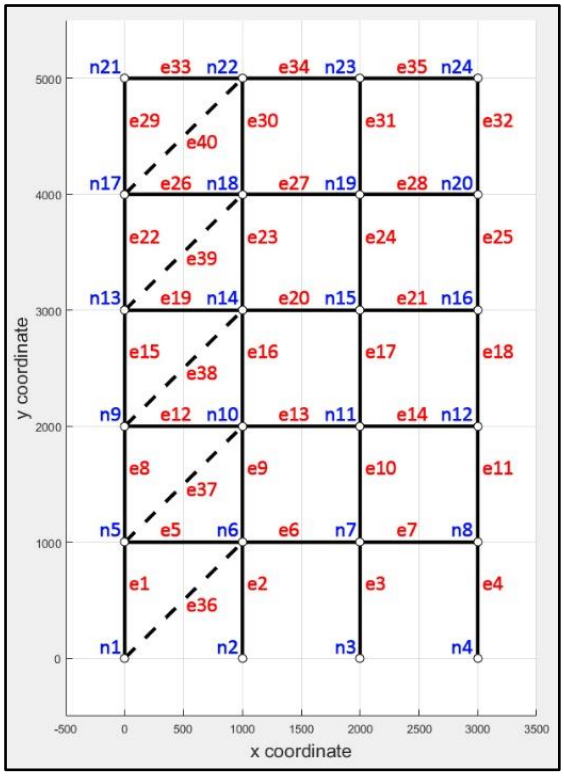


Figure 1.1 Nodes and elements numbering convention.

1.3 Possible positions of cross member on a storey

Since the structure has three bays, there are six possible positions of the cross member on each storey. They are generated in a matrix called ‘config’ and are numbered as shown:

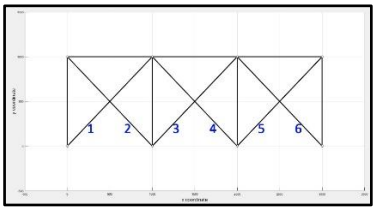


Figure 1.2 Numbering of possible positions of cross member.

2.0 Question 1 – Truss stability

The stability of a truss is related to its number of redundancies, which can be calculated using:

$$Number\ of\ redundancies = m + 2 - 2j,$$

where m is the number of members in the truss, r is the number of reaction components and j is the number of pin joints. If the number of redundancies is smaller than zero, the truss is unstable. In the second section of the main script, six truss structures with increasing number of cross members are generated, as shown in figure 2.1.

Redundancy	rcond
-5	9.37×10^{-35}
-4	7.97×10^{-35}
-3	4.66×10^{-35}
-2	6.93×10^{-35}
-1	6.93×10^{-35}
0	1.63×10^{-4}

Table 2.1 The value of rcond of the matrix K_{FF}.

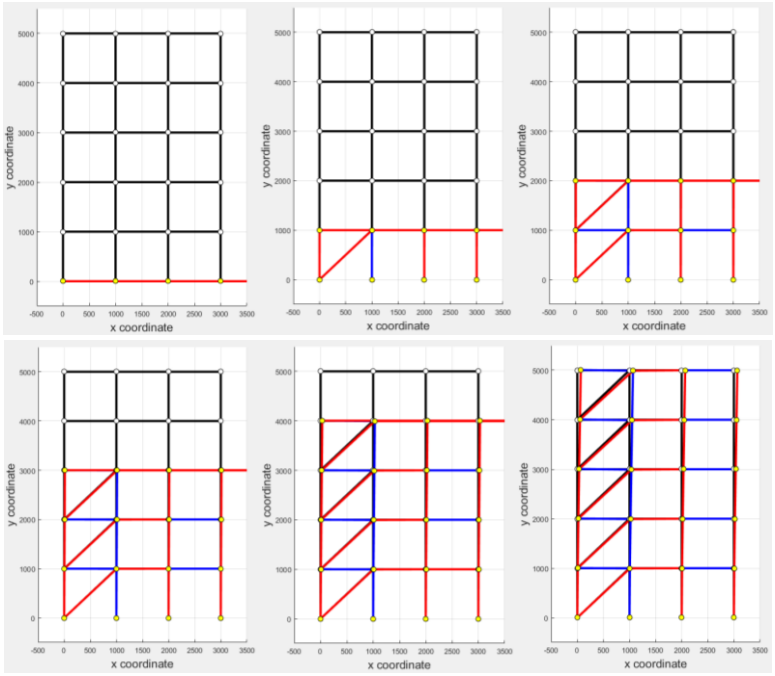


Figure 2.1 Increasing number of cross members.

Figure 2.1 shows that the truss structure collapses when its number of redundancies is smaller than zero. This is reflected in the values of rcond of the matrix K_{FF} in table 2.1, which shows that only the matrix of the last truss structure (number of redundancies equals to zero) is well-conditioned i.e. the value of rcond is not near 10^{-15}. This means that a minimum of 5 cross members are required to ensure the stability of the truss structure.

A truss structure with five cross members is now generated; instead of having one cross member on each storey, there are two cross members on the fourth storey and none on the fifth storey. The number of redundancies of the structure is equal to zero.

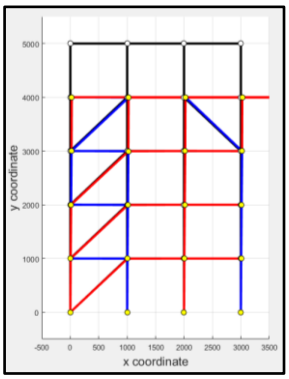


Figure 2.2 Structure with a number of redundancies of zero, but still collapses.

For this structure, the value of rcond of the matrix that is 'inverted' is 6.96×10^{-35}. This proves that even though the number of redundancies is equals to or larger than zero, the structure can be unstable. In other words, the position of the cross-members is also important.

In the case when the number of redundancies equals to or larger than zero, the stability can be easily examined using the method of inspection. Since not all the reactions acting on the structure are parallel and concurrent, the truss is externally stable. As for the internal stability, the arrangement of the elements is examined based on the fact that a pin-jointed triangle is stable, and that the basic triangle can be extended whilst maintaining the stability by adding two new elements and a new node. In addition, if a new element is connected to two existing nodes, the stability is not affected and the redundancy increases by one. The pin-jointed quadrilateral on the fifth storey is unstable, hence only that storey has collapsed. Therefore, the stability of the truss structure can only be maintained if there is at least one cross member on each storey.

3.0 Question 2 & 3 – Possible arrangement of cross members

As shown in section 1.3, there are six possible positions of cross member on each storey. If there is only one cross member on each storey, the number of possible permutations is:

$$Number\ of\ possible\ permutations = (2 \times Number\ of\ bays)^{Number\ of\ storeys}.$$

Therefore, for a structure with three bays and five storeys, there are 6^5=7776 possible permutations. Each possible permutation can be represented by a vector (called 'arrangement' in the scripts) with 5 elements, where the numbers (one to six as in section 1.3) in the vector are the positions of the cross member on each storey starting from ground level. For example, the last truss structure in figure 2.1 has an 'arrangement' of [1,1,1,1,1].

When generating 'arrangement', the characteristics of numbers in base six can be exploited, where the numbers in base six are between and including zero and five. If every numbers from 0 to 7775 are

converted to that of base six, and each number is then added by one, then every possible permutation of 'arrangement' are covered. For example, the number 7775 in base six is 55555. After adding one to each number, the 'arrangement' is [6,6,6,6,6].

Using a for loop, every 'arrangement' of the cross members are considered. The lateral sway for each arrangement (which is the lateral sway of the top left node) can be found using the stiffness method. After the preliminary steps where all the necessary inputs (such as the nodal coordinates, the element connectivity, the restrained dofs and the applied forces) are defined, the stiffness matrix is constructed by adding the 4x4 matrix shown below to the corresponding dofs of the two connected nodes for each element:

$$\frac{EA}{L}\begin{pmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{pmatrix},$$

where c=cos(α), s=sin(α), and α is the angle of the element to the horizontal. The unknown dofs and reactions are then found using the equations:

$$u_F = {K_{FF}}^{-1}(f_F - K_{FR}u_R) \text{ ----- (Eq. 1)}$$

$$f_R = K_{RF}u_F + K_{RR}u_R. \text{ ----- (Eq. 2)}$$

These steps are done using the method 'Solve' as described in section 1.1.

Equilibrium checks, both horizontally and vertically, are also done within the for loop to ensure the accuracy of the calculations. As described in section 1.2, the method 'EquilibriumCheck' outputs ones if the check is passed and zeros otherwise. These outputs are stored, so if zero(s) is/are present, the equilibrium check is failed.

3.1 Optimal arrangement of cross members

It is found that the smallest possible lateral sway is 15.8968mm, and the 'arrangement' of cross members is [1,3,6,4,2], which is shown in figure 3.1. The axial loads in each elements are also displayed. Elements in compression are shown in blue, while elements in tension are shown in red.

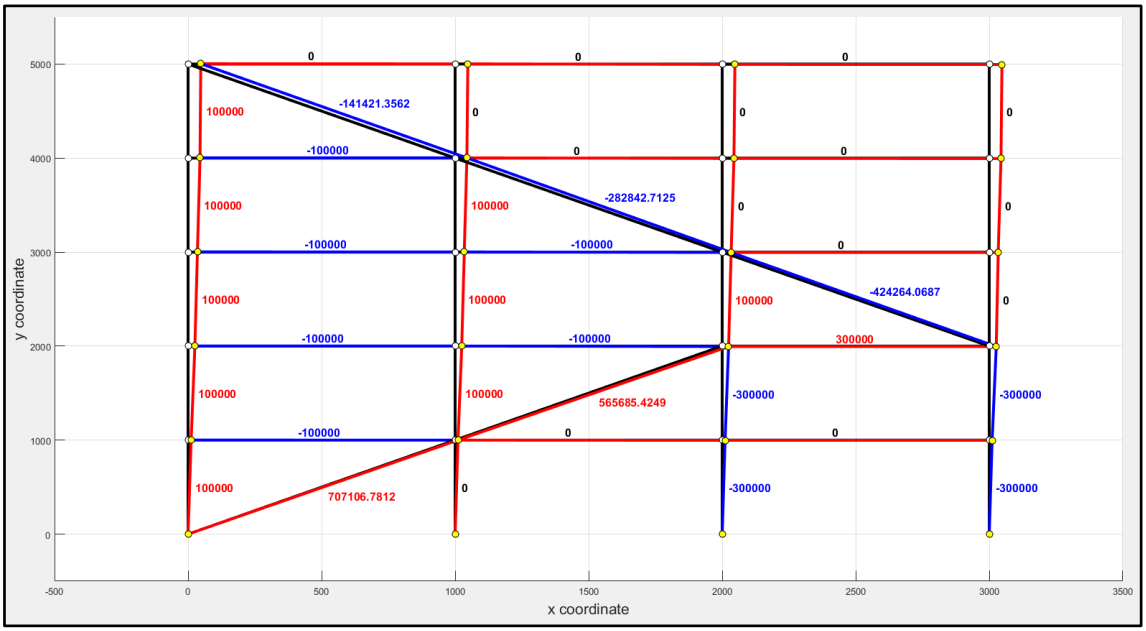


Figure 3.1 Optimal arrangement of cross members (Amplification factor of 3)

3.2 Least-optimal arrangement of cross members

It is found that the largest possible lateral sway is 67.3356mm, and the 'arrangement' of cross members is [5,5,5,5,5], which is shown in figure 3.2. The axial loads in each elements are also displayed. Elements in compression are shown in blue, while elements in tension are shown in red.

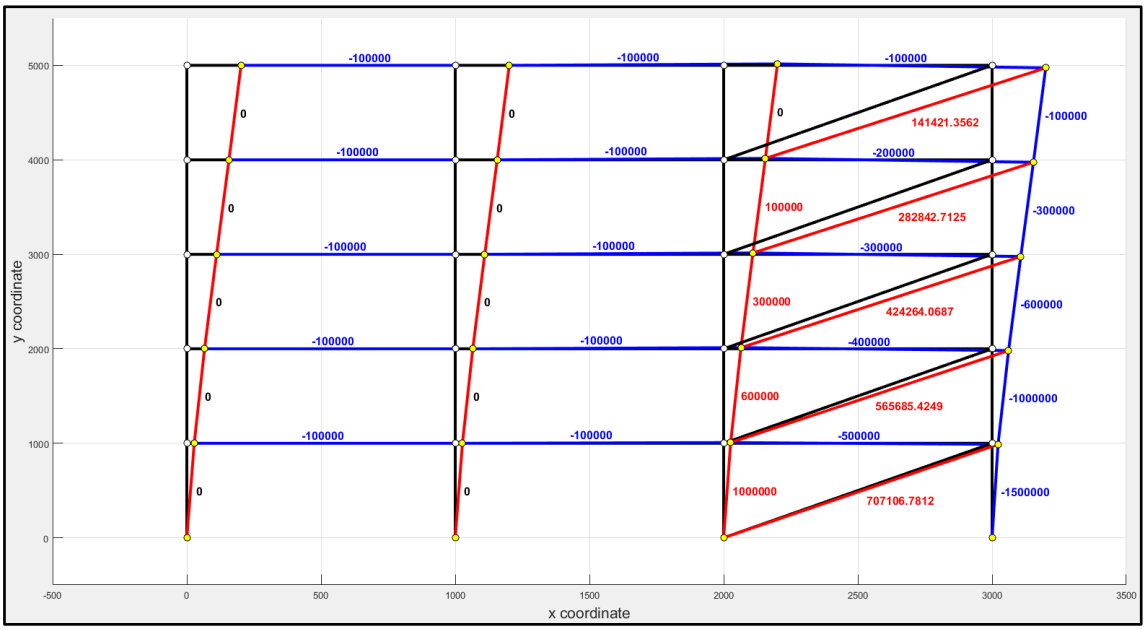


Figure 3.2 Least-optimal arrangement of cross members (Amplification factor of 3)

3.3 Comparison of optimal and least-optimal arrangements

The ability of both arrangements to resist lateral sway can be explained using the concept of critical path, where the critical path is the shortest distance between the top point of horizontal load application and the ground.

In figure 3.1, the cross members of the optimal structure are arranged such that the critical path is the shortest. The top three cross members in compression are continuous, allowing effective transfer of horizontal loads to the ground. On the other hand, the bottom two cross members are in tension, resisting the moment about node 4. All this contribute to the smaller lateral sway. As for the least-optimal structure, the longer critical path and the discontinuity between the cross members results in larger lateral sway.

4.0 Question 4 – Frame structure

Frames do not need cross members to be stable because the members are continuous at the joints, i.e. they can transfer rotational load, resulting in a number of redundancies much larger than zero.

For a Timoshenko beam element, the total transverse deformation is the sum of that due to bending and that due to shear. The deformation due to shear is dependent on the shear ratio ϕ. If the members are assumed to be slender ($L \to \infty$), the shear ratio ϕ and so the deformation due to shear are almost zero. Hence, the total deformation is only due to bending and Euler-Bernoulli beam element stiffness matrix can be used.

The procedure to solve for the lateral sway in a frame is mostly similar to that in a truss; the equations 1 and 2 in section 3.0 are used. For frame structures, three dofs are needed per node instead of two, where the third dof is the rotational displacement. This means that a 6x6 matrix is added for each element when constructing the stiffness matrix:

$$\begin{bmatrix} \frac{EA}{L}c^2+\frac{12EI}{L^3}s^2 & \frac{EA}{L}cs-\frac{12EI}{L^3}cs & -\frac{6EI}{L^2}s & -\frac{EA}{L}c^2-\frac{12EI}{L^3}s^2 & -\frac{EA}{L}cs+\frac{12EI}{L^3}cs & -\frac{6EI}{L^2}s \\ & \frac{EA}{L}s^2+\frac{12EI}{L^3}c^2 & \frac{6EI}{L^2}c & -\frac{EA}{L}cs+\frac{12EI}{L^3}cs & -\frac{EA}{L}s^2-\frac{12EI}{L^3}c^2 & \frac{6EI}{L^2}c \\ & & \frac{4EI}{L} & \frac{6EI}{L^2}s & -\frac{6EI}{L^2}c & \frac{2EI}{L} \\ & sym & & \frac{EA}{L}c^2+\frac{12EI}{L^3}s^2 & \frac{EA}{L}cs-\frac{12EI}{L^3}cs & \frac{6EI}{L^2}s \\ & & & & \frac{EA}{L}s^2+\frac{12EI}{L^3}c^2 & -\frac{6EI}{L^2}c \\ & & & & & \frac{4EI}{L} \end{bmatrix},$$

where c=cos(α), s=sin(α), and α is the angle of the element to the horizontal. The second moment of area, I of the cross section is required to construct the stiffness matrix K. For a circular cross section:

$$Second\ moment\ of\ area, I = \frac{\pi}{4}r^4, \text{ ----- (Eq. 3)}$$

where r is the radius of the circle.

It is found that the lateral sway at the top left node of the frame structure is 1082.04mm, and the deformed shape of the structure is shown in figure 4.1. The axial loads in each elements are also displayed. Elements in compression are shown in blue, while elements in tension are shown in red.

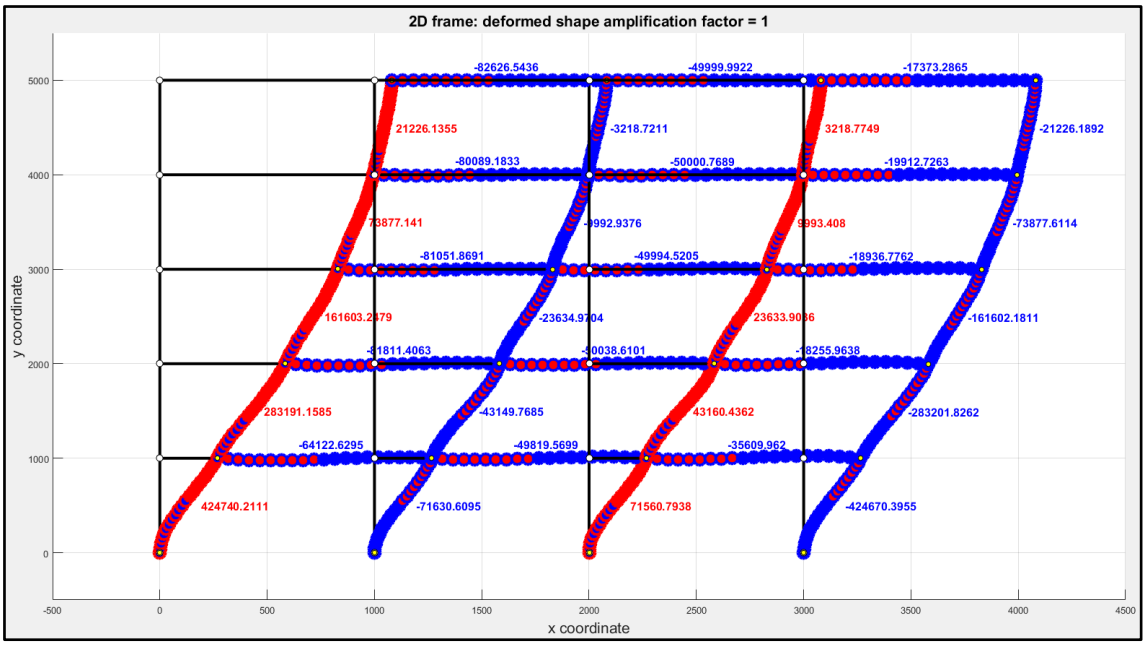


Figure 4.1 Deformed shape of the frame structure.

The lateral sway in the frame structure is much larger than that in the truss structure. This is because the only way for the frame structure to resist horizontal load is through the bending resistance of the elements.

The relative efficiency is dependent on the lateral sway and the total volume of steel used. The larger the lateral sway, the less efficient the structure is; likewise, the larger the total volume, the less efficient the structure is.

Structure	Lateral Sway [mm]	Total volume [mm^3]
Optimally arranged truss	15.8968	$8.26x10^7$
Least-optimally arranged truss	67.3356	$8.26x10^7$
Frame	1082.04	$6.87x10^7$

Table 4.1 Relative efficiencies

Even though the frame structure uses less material, due to the significant increase in lateral sway, both the truss structures are relatively more efficient than the frame structure in resisting lateral sway. A better comparison of efficiency is discussed in section 5.0.

5.0 Question 5 – Minimum radius and volume of steel

The structures can be made more economical by reducing the radius of the members. To find the minimum cross-sectional radius r that results in exactly 25mm of lateral sway, two function files named 'TrussSway' and 'FrameSway' are written. Using the inputs and the classdef 'TRUSS' or 'FRAME', the unknown dofs are solved using the stiffness method. The output value is the lateral sway at the point of application of the top-most load minus 25mm. So, if the output is zero, the minimum radius is found. This allows the use of the iterative root-finding function 'fzero'. The volume of steel used for each structure is also calculated, using the method 'Volume'. The results are shown in table 5.1.

Structure	Minimum radius [mm]	Volume of steel [m^3]
Optimally arranged truss	19.94	0.0525
Least-optimally arranged truss	41.03	0.2225
Frame	64.63	0.4594

Table 5.1 Minimum radius and volume of steel.

The structure that uses lower volume of steel is more economical since the price is directly proportional to the volume of steel used. The optimally arranged truss structure is the most economical, followed by least-optimally arranged truss structure and frame structure. Besides, it is usually more expensive to produce fixed joints (joints that can transfer rotational loads). This makes the frame structure even less economical.

6.0 Question 6 – Euler critical buckling load and margin of safety

Although reducing the radius can cut down the cost, it makes the members slenderer. This means that if the members are in compression, they are more susceptible to buckling. The axial compression resistance of slender members corresponds to the Euler critical buckling load:

$$P_{cr} = EI\frac{\pi^2}{l^2},$$

where E is the Young's Modulus of steel, I is the second moment of area (shown in equation 3) and l is the effective buckling length of the member. For the truss structures, since all members are pin-jointed at both ends, the effective buckling length is the length of the member. Whereas for the frame structure, the members are clamped at both ends, hence the effective buckling length is half the length of the member. This is because if the member is pin-jointed at both ends, buckling is more likely to occur. Therefore, the higher effective length decreases the axial compression resistance.

After the Euler critical buckling load is found, the margin of safety is calculated. The margin of safety is defined as:

$$Margin\ of\ safety = \frac{P_{cr}}{N} - 1,$$

where N is the actual axial load in the member. If the margin of safety of a member is smaller than zero, i.e. the actual axial compressive load is larger than Euler critical buckling load, then the member is at risk of buckling. However, since only members in compression are vulnerable to buckling, if the member is in tension (axial load greater than zero), then the margin of safety is not calculated.

The calculations of Euler critical buckling loads and the margins of safety for each element can be done using the method 'Buckling Check'. This requires the axial load to be calculated first, using the method 'AxialLoad'. The structures are then plotted using the method 'PlotBuckling'. The elements in red are compressive members at risk of buckling; the elements in blue are compressive members not at risk of buckling; the elements in black are tensile members. The values of margin of safety for each member are also shown in the figures.

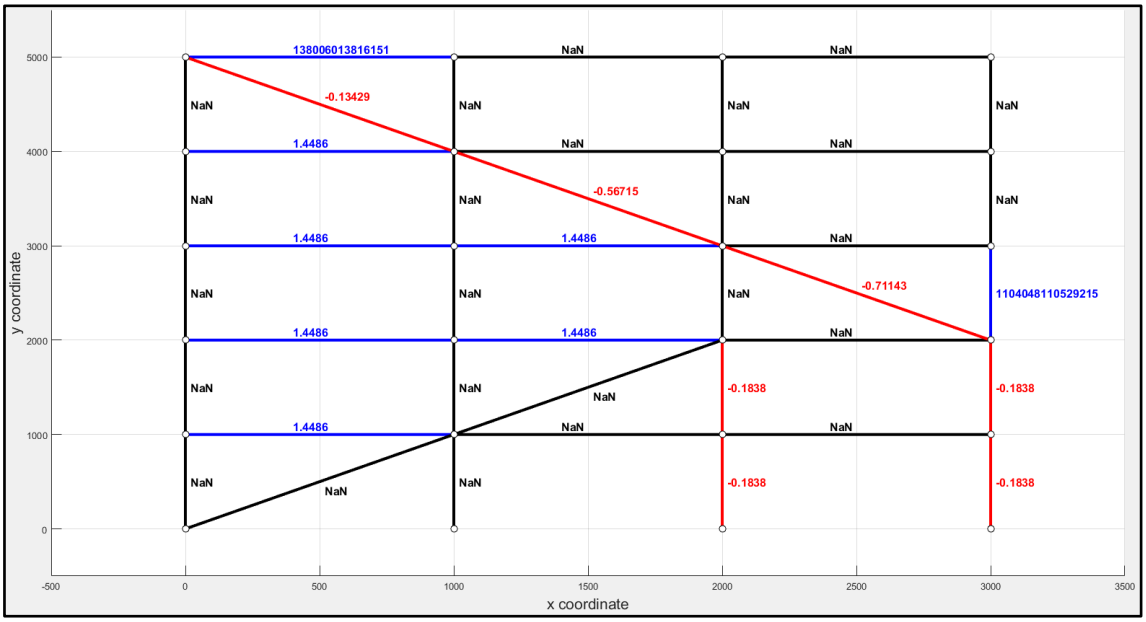


Figure 6.1 Members at risk of buckling in the optimally arranged truss structure.

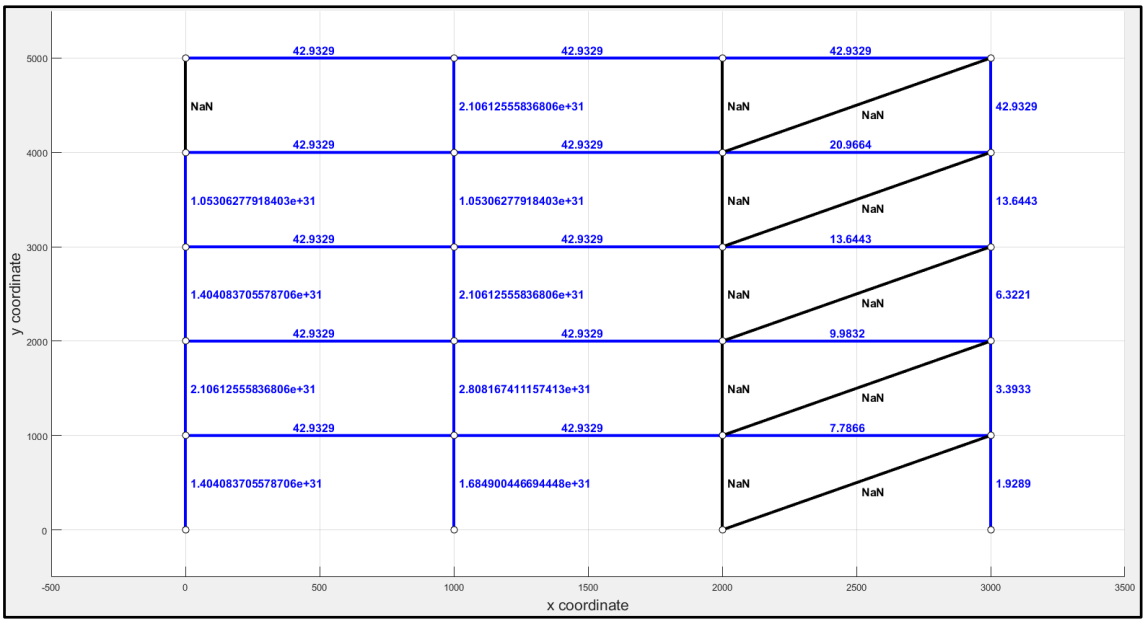


Figure 6.2 Members at risk of buckling in the least-optimally arranged truss structure.

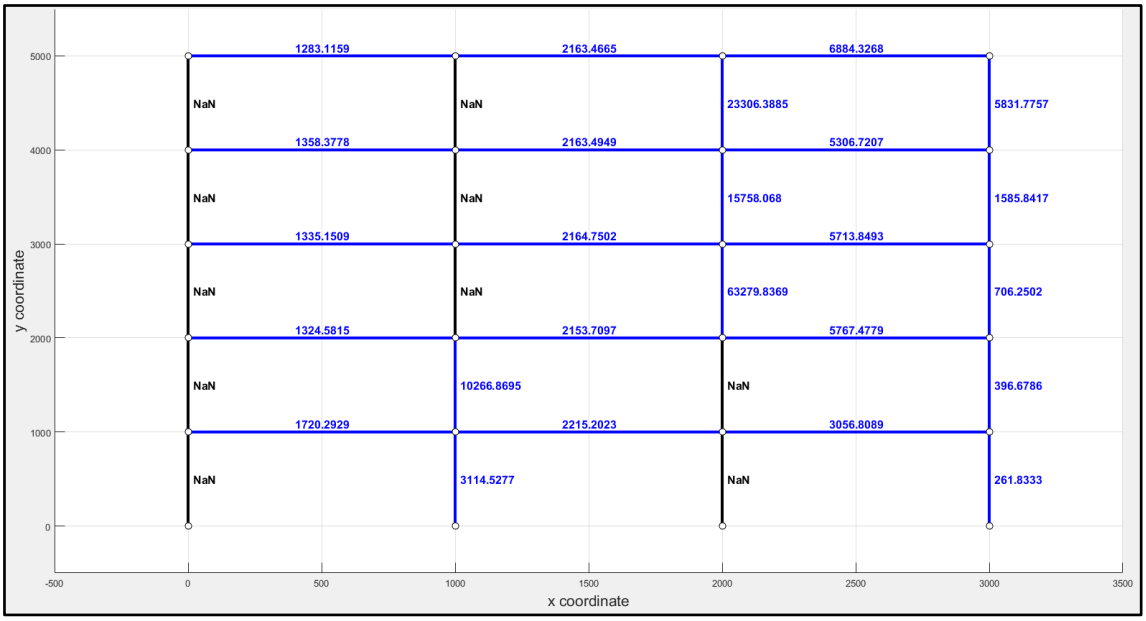


Figure 6.3 Members at risk of buckling in the frame structure.

The margin of safety of the overall structure is defined at its weakest point; it is the smallest value. The margin of safety for each of these structures are shown in table 6.1.

Structure	Margin of safety
Optimally arranged truss	-0.711
Least-optimally arranged truss	1.929
Frame	261.833

Table 6.1 Margin of safety of each structure.

The members in the optimally arranged truss structure are not only thinner, but also experiencing larger compressive force compared to other two structures. This is shown in figures 3.1, 3.2 and 4.1. Besides, compared to the frame structure, the effective buckling length is longer. Therefore, the margin of safety

for the optimally arranged truss structure is the smallest, and is the only structure with members at risk of buckling.

7.0 Further optimisation

To prevent buckling, the radii of the members at risk of buckling should be increased. For each of these members, the radius such that the margin of safety is zero is found. This is done using the method 'BucklingRadius'. Since the radii of some members have increased, the overall lateral sway of the structure decreases. Hence, the same optimisation procedure as in section 5.0 is performed such that the lateral sway is 25mm. However, only the radii of the members not at risk of buckling are decreased. The radii and the margin of safety for each element are shown in figures 7.1 and 7.2. It is seen that the structure is safe from buckling, and the volume of steel used is $0.0481m^3$, which is even smaller and more economical.

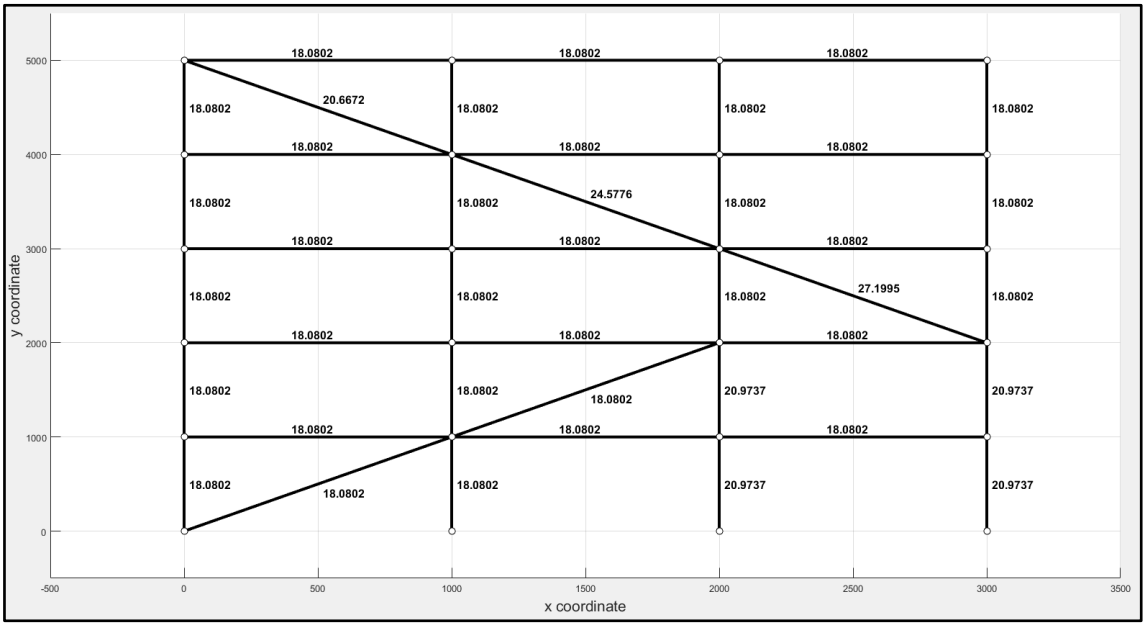


Figure 7.1 Radius of each element in the optimised truss structure.

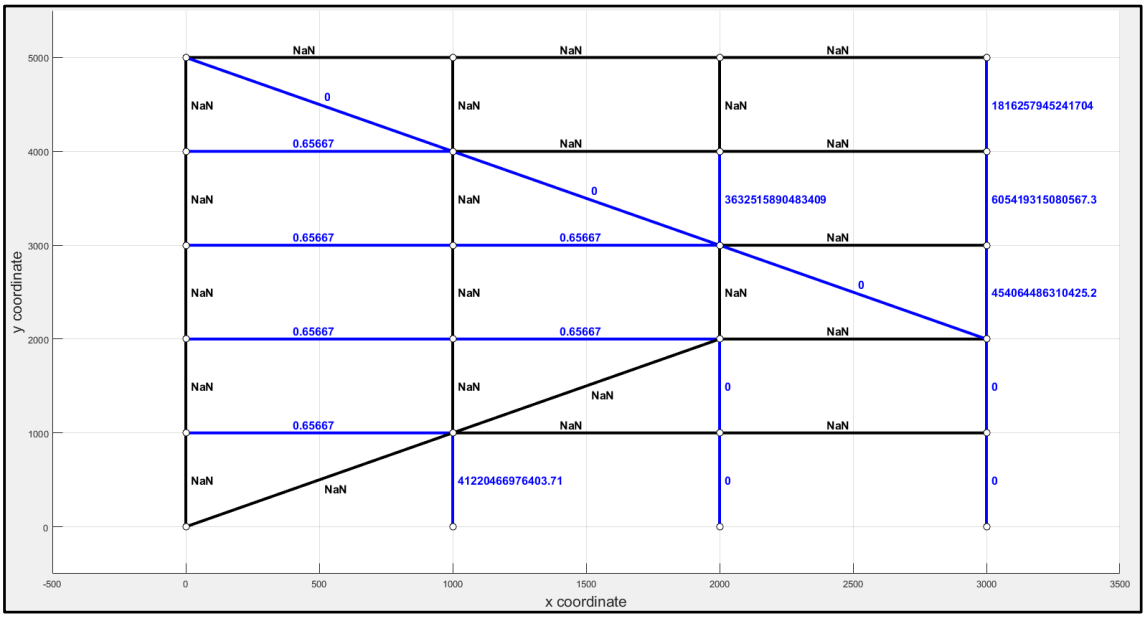


Figure 7.2 Members at risk of buckling in the optimised truss structure.

8.0 Conclusion

A truss structure is unstable if the number of redundancies is smaller than zero. Otherwise, the method of inspection can be used: a pin-jointed triangle is stable, and the basic triangle can be extended whilst maintaining the stability by adding two new elements and a new node.

All possible permutations are tested and the optimal and least-optimal arrangements are found. The optimally arranged truss structure has a shorter and more effective critical path that transfers the horizontal load to the ground, resulting in a smaller lateral sway.

The truss structures are then compared to a frame structure. Even though frame structure uses less steel due to the absence of cross members, the significantly larger lateral sway makes the truss structure a better choice in terms of its efficiency. This is further proven in section 5.0 where the optimally arranged truss structure requires the least steel to allow the same lateral sway.

However, because slender members in compression are susceptible to buckling, a check is done in section 6.0. The optimally arranged truss structure is at risk of buckling. The margin of safety for the optimally arranged truss structure is the smallest due to the thinner members and the relatively larger compressive force experienced compared to other two structures.

Further optimisation is done in section 7.0, where the radii such that the margin of safety is zero is found. For a structure with 25mm lateral sway, the volume of steel used is reduced to $0.0481m^3$.